

Low-Rank Architectures and Deep Optimization

Kernel, mean-field, and feature-learning perspectives

Janis

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Neural networks: scaling laws and depth vs. width

The practical problem

- We want deep nets to approximate functions or fit data.
- **Optimization** and **architecture** interact in poorly understood ways.
- Classical wisdom: “go wider and deeper.” Large-scale evidence refines that.

Scaling laws (Kaplan et al. / OpenAI-style)

- Test loss often scales as a **power law** in compute, data, and model size.
- With matched compute and data, **depth and width trade**. Many splits give similar loss if parameters and training budget match.
- What matters first is **scale** (parameters, steps, tokens). Not a unique depth recipe.
- Takeaway: need **structure** (rank, bottlenecks, RF). Not only more MLP layers.

Optimization tricks—and what we still do not understand

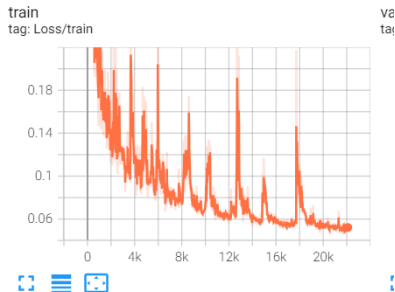
What practitioners use

- **SGD** + momentum: damp oscillations. Faster progress when ill-conditioned.
- **Adaptive**: RMSProp, **Adam** / AdamW. Per-parameter LR. Often the default.
- **Schedules**: warmup, cosine decay, step decay. Often as important as architecture.

Gap

Sharp guarantees for Adam on deep nonconvex losses are limited. Behavior is **tuned** in practice.

Adam can look “unstable”



Loss can **wobble**: sharp oscillations even when the average trend improves. Sensitive to LR, noise, curvature. Motivates **gentler** landscapes (e.g. low-rank RF).

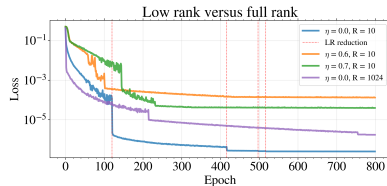
MMNN (Zhang et al.): low rank + frozen features $\Rightarrow$ very low test error

Structured MMNN

- Each layer: $\mathbf{h} = \mathbf{A}\sigma(\mathbf{W}\mathbf{x} + \mathbf{b}) + \mathbf{c}$.
- $\mathbf{W}, \mathbf{b}$ frozen (RF). Train $\mathbf{A}, \mathbf{c}$ only. Multi-component rank between layers.
- Same param budget as FC. Often **much smaller** test MSE on oscillatory targets.

Reported test MSE (SIAM MMNN paper, excerpt)

Setting	MMNN (S1)	MMNN (S2)	FCNN
Strategy comp. (osc. 1D)	2.0×10^{-5}	4.3×10^{-5}	—
Best MMNN row	1.4×10^{-5}	—	—
f_1 (1D oscillatory)	2.5×10^{-6}	2.1×10^{-6}	1.7×10^{-4}
f_2 (2D oscillatory)	4.6×10^{-6}	6.2×10^{-6}	3.3×10^{-5}



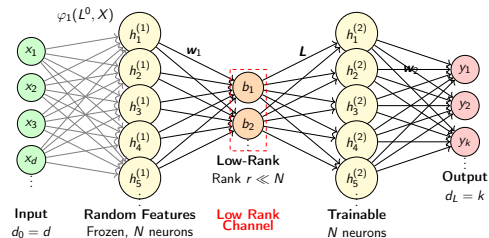
rank $r=10$ vs full-rank ($r=1024$) on $y=\cos(2\pi x)$, $x \in [-1, 1]$, 3 layers, width 1024, $n=5000$, batch=4, SGD lr 0.01 with rec (momentum $\eta \in \{0, 0.3, 0.6, 0.7\}$) and one full-rank ($\eta=0$). Red bars: SGD lr reductions for $\eta=0$.

Message

Low rank + excellent MSE. Training $\mathbf{A}$ is often **more benign** than full FC at comparable width.
MNIST-class: RF-LR/MMNN near a wide MLP. **Far fewer** trainable weights (Zhang et al.).

Design principle

- Keep a large random feature bank.
- Train only through a small rank- r bottleneck.
- Preserve expressive behavior while keeping analysis tractable.
- Make channel specialization visible during training.



Main thesis and roadmap

From scaling laws, optimizers, and MMNN to this talk

- **Scaling laws** (Kaplan et al.): matched compute/data $\Rightarrow$ loss tracks scale. Depth/width **trade**.
- We stress **structure** (rank, frozen RF). It reshapes optimization.
- **Optimization**: SGD/momentum vs. Adam. Schedules matter. **Wobbles** $\Rightarrow$ conditioning and spectra.
- **MMNN** (Zhang et al.): frozen W , train A . **Low test MSE** vs. FC. Motivates RF-LR theory.

Core claim

- Low rank is not only compression.
- It isolates optimization-relevant scales.

Outcome

- One framework: depth, rank, spectra, features.
- NTK and training dynamics together.

Program

Theorems: MNIST-style efficiency. Oscillatory regression. Kernel recursions. Step-size stability. Matches MMNN / RF-LR.

NTK at a glance: settled limit, unfinished business

Mostly resolved (infinite width)

- Sequential limit $\Rightarrow$ **lazy training**. Deterministic NTK (Jacot; Chizat).
- Explicit recursions. Empirical kernel convergence. Coherent **kernel picture**.

Not the full optimization story

- Finite width: feature learning. **Time-dependent** kernel.
- NTK is **blunt** for end-to-end SGD.
- Still useful for the **initial** curve, conditioning, spectra at large finite width.

Large but finite width

- **Feynman diagrams**: Guillen, Misof & Gerken, arXiv:2508.11522.
- Systematic **finite-width** NTK corrections (dNTK/ddNTK). Beyond Gaussian stats.
- Recursions for **early** dynamics.
- **Spectral theory**: Benigni & Paquette, arXiv:2508.20036.
- NTK **eigenvalue distribution**. Quadratic scalings (sample–dimension–width). Near interpolation.

Takeaway: refined NTK tools help the **opening** of training and stability. Before mean-field / feature learning.

Neural Tangent Kernel (NTK)

Lazy training and the kernel

- Infinite-width (sequential) limit: training linearizes.
- Optimization governed by a fixed Gram matrix and its spectrum.
- RF-LR NTK: explicit layerwise recursion. Visible $1/r$ per bottleneck. Rank at leading order.

What we read off from the NTK

- Local convergence. Step-size stability. E.g. $\lambda_{\max}$, conditioning of centered kernel.
- Depth scaling of correlations. Kernel geometry. Edge-of-chaos alignment across layers.
- RKHS comparison: expressivity vs optimization geometry. When class matches shallow ReLU despite low rank.

High-dimensional and concentration

- Spherical / equicorrelated data: simpler spectra.

Mean-field viewpoint

Beyond a fixed kernel

- Mean-field: evolving weight distributions as width grows. Measures or particles.
- Training need not stay lazy.
- Deep RF-LR: tractable limiting flow.
- Frozen RF: rich mixing directions. Low-rank readouts train.

What the mean-field analysis targets

- Well-posed limiting dynamics. Finite-width error control toward the limit.
- **If** dynamics converge: limit can be a **global** minimizer of population loss. i.i.d. init.
- Avoids degenerate collapse from full-rank multilayer mean-field. Under RF-LR constraints.

Feature learning

- Specialization. Log-ratio growth. Structured internal features.

Theorem 1: Infinite-Width NTK Recursion

Statement

- Setup: sequential infinite-width limit of the RF-LR NTK.
- Object: recursion for the mean kernel across layers.

$$\begin{aligned}\Theta^{(1)} &= 1 + \Sigma^{(1)} \\ \Theta^{(\ell)} &= 1 + \frac{1}{r} \Theta^{(\ell-1)} \dot{\Sigma}^{(\ell)} + \frac{1}{r} \Sigma^{(\ell)}\end{aligned}$$

Interpretation

The recursion makes the low-rank contribution explicit.
Each bottleneck contributes a visible factor $1/r$.
Rank therefore enters the kernel at leading order.
This is the basis for the later depth and conditioning results.

Theorem 2: Depth Scaling of the Proxy Kernel

Statement

- Setup: proxy kernel along deep RF-LR layers.
- Object: correlation alignment and diagonal-off-diagonal gap.

$$1 - \rho_k = \Theta(k^{-2})$$

$$\Theta_{\text{diag}}^{(k)} - \Theta_{\text{off}}^{(k)} = \Theta\left(\frac{1}{rk}\right)$$

Interpretation

Depth drives the proxy kernel toward a nearly constant regime.

The informative gap decays at rate $1/(rk)$.

Rank moderates, but does not remove, this collapse.

The theorem quantifies the depth-rank tradeoff in the kernel regime.

Theorem 3: Conditioning Regimes

Statement

- Setup: condition number of the centered RF-LR kernel.
- Compare generic data with equicorrelated and spherical regimes.

$$\kappa \geq \Omega(rL)$$

$$\kappa_{\perp} = 1, \quad \kappa_{\perp} = 1 + o(1)$$

Interpretation

In generic geometries, conditioning worsens with depth and rank.
Equicorrelated and spherical regimes remain well conditioned.
The optimization picture is therefore geometry-dependent.
Architecture alone does not determine conditioning.

Comparison table

	Fully trained nets	RF-LR / MMNN
Trainable	all layers	readouts, biases
Scaling	depth and EOC	depth and rank
Kernel size	can grow	saturates
Centered gap	correlation driven	$\asymp 1/(rL)$

Message

The optimization question is the same in both settings.
The RF-LR model makes the relevant scales explicit.
Depth, rank, and conditioning can be tracked directly.
This is why it is useful as a reference model.

Theorem 4: Low Rank Is Enough for the RKHS

Statement

- Setup: three-layer mean RF-LR kernel with isotropic random features.
- Randomness changes coefficients, but not the endpoint exponent.
- Claim: the induced RKHS equals the shallow ReLU RKHS.

$$\mathcal{H}_{\text{RF-LR}} = \mathcal{H}_{\text{ReLU}}$$

Interpretation

The random low-rank bottleneck does not change the RKHS exponent.
It modifies the coefficient, not the functional class.
The gain therefore does not come from a larger RKHS.
It comes from conditioning, spectra, and training geometry.

Theorem 5: Mean-Field Well-Posedness and Global Minimizers

Statement

- Setup: mean-field dynamics for deep RF-LR training.
- The mean-field flow is well posed.
- If the dynamics converges, the limit is globally optimal.

Interpretation

The result holds under standard i.i.d. initialization.

Frozen random features preserve the required richness across depth.

This avoids the collapse mechanism present in full-rank multilayer theory.

It provides the dynamic counterpart to the kernel results.

Theorem 6: Log-Ratio Growth and Channel Dominance

Statement

- Setup: compare two channels at a fixed point x_0 .
- Observable: the log-ratio $R_{12}(t, x_0)$.

$$\partial_t R_{12}(t, x_0) \geq 0$$

Interpretation

Log-ratios quantify channel dominance at a fixed location.
Positive growth implies persistence and amplification of dominance.
The specialization mechanism is therefore dynamical, not only visual.
This gives a rigorous entry point into feature learning.

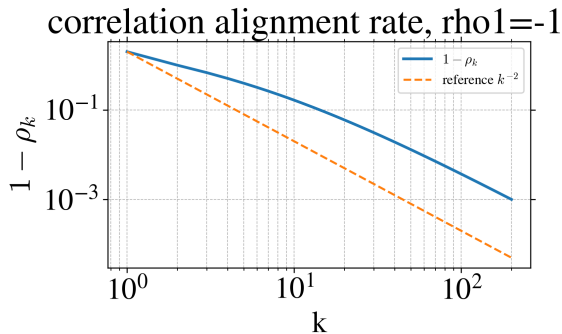
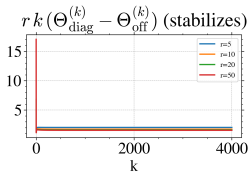
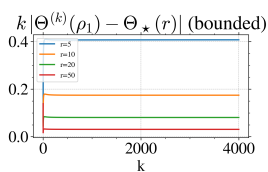
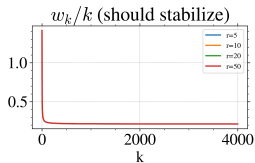
Theory

- Explicit scaling laws. Not only heuristic depth–rank talk.
- Weaker assumptions than earlier drafts.

Diagnostics

- Expressivity split from conditioning and optimization.
- Observables for feature learning: log-ratios, partials, symmetry.

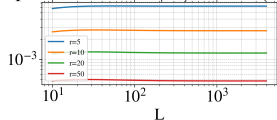
Depth and Rank in Kernel Geometry



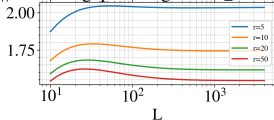
- The empirical curves follow the predicted scaling law.
- Correlations align across depth.
- Depth flattens the kernel geometry.
- Rank controls the decay of the informative gap.

Further Evidence for Depth Scaling

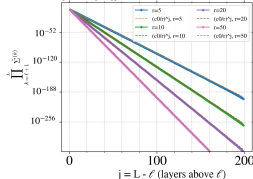
spike saturation: $L |\lambda_1/n - \Theta_\star(r)|/n$



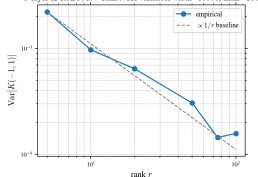
gap scaling: $r L \lambda_\perp$



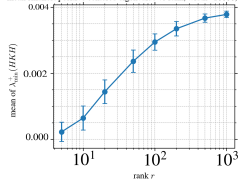
exponential depth suppression, ReLU EOC, $x=1.0, x'=1.0, L=200, c_0=0.5$



3-layer (2-ReLU) low-rank NTK variance, width=16000, trials=100

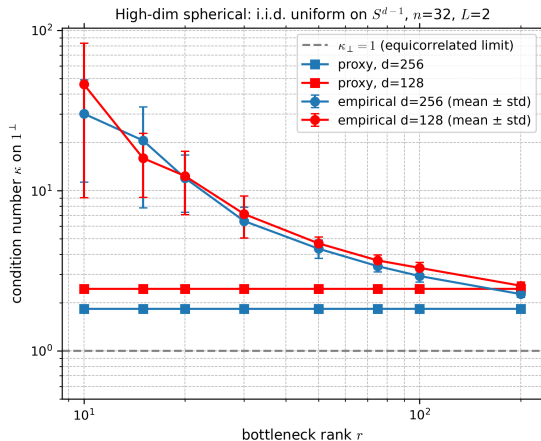
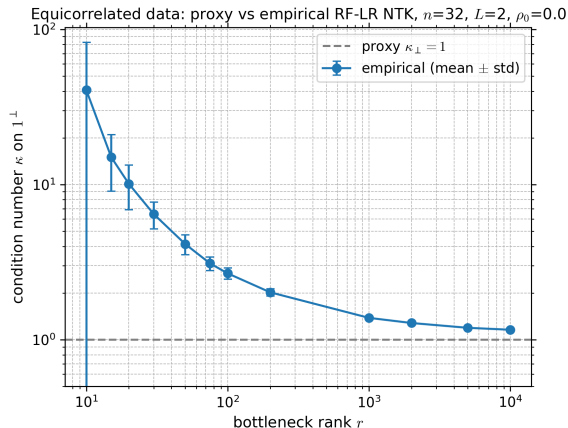


mean smallest positive centered eigenvalue vs rank, width=16000.



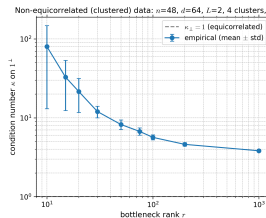
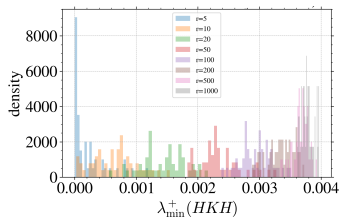
- Four observables support the same scaling law.
- Gap decay, variance decay, and spectral behavior are consistent.
- Finite-rank corrections remain visible numerically.
- The agreement is not confined to a single statistic.

Conditioning Is the Optimization Bottleneck



- The deterministic proxy tracks the empirical condition number well.
- Data geometry drives the conditioning behavior.
- Benign regimes remain stable across depth.

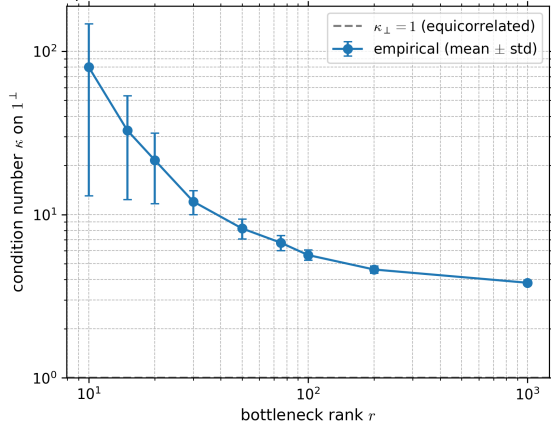
Smallest Eigenvalue and Difficult Regimes



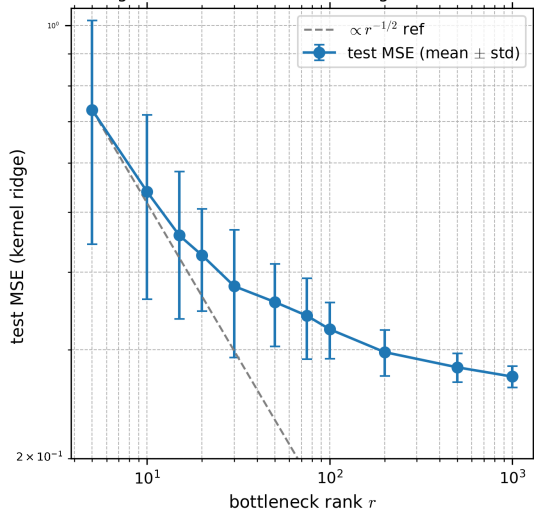
- The lower edge of the spectrum controls the hardest cases.
- Small eigenvalues slow convergence and reduce stability.
- The histogram makes this concentration visible.
- Low rank does not erase difficult geometry.

Rank Helps, But Not Uniformly

Non-equicorrelated (clustered) data: $n=48$, $d=64$, $L=2$, 4 clusters,

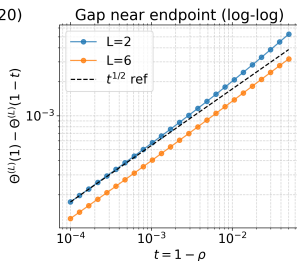
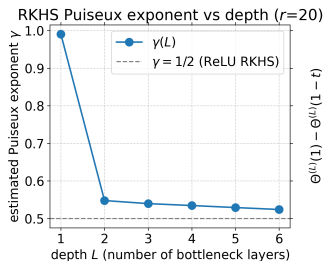


Kernel regression: RF-LR NTK, linear target, $n=64$, $d=32$, $L=2$



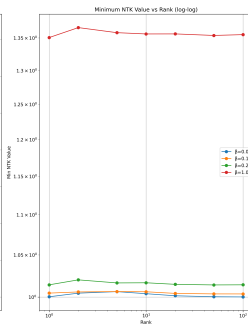
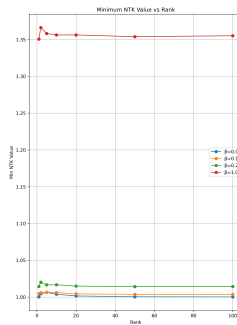
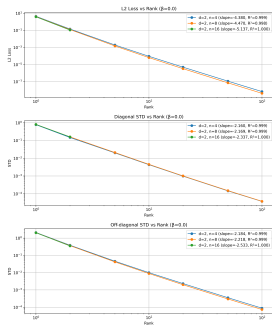
• Low rank is not a universal remedy

Expressivity Is Not the Whole Story



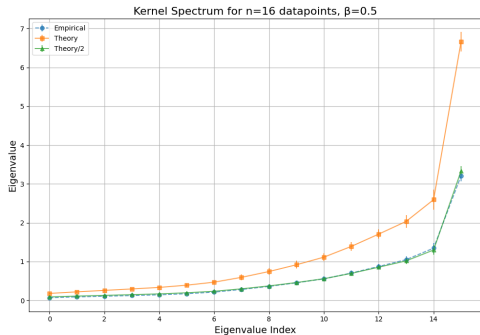
- The RKHS can stay the same.
- The optimization geometry can still change a lot.
- Learned features are reached differently.
- Expressivity alone is not the explanation.

Early Empirical NTK Evidence



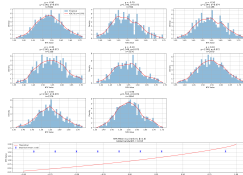
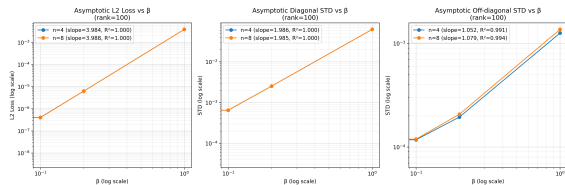
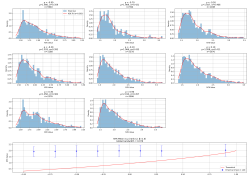
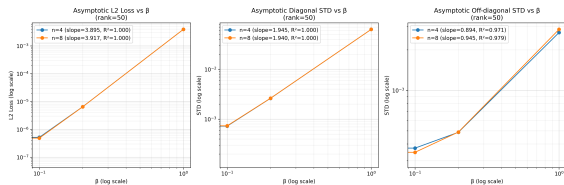
- Rank changes both kernel magnitude and spectral spread.
- The effect is already structured in the early experiments.
- These plots motivated the later theory.
- The subsequent results explain this empirical pattern.

Spectrum Comparison



- The MMNN spectrum approaches the MLP benchmark systematically.
- Increasing rank sharpens concentration.
- The comparison persists across scales.
- This suggests a genuine spectral correspondence.

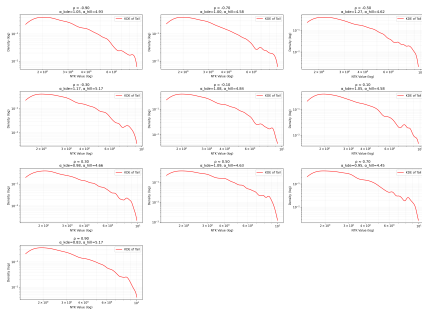
Kernel Statistics Beyond the Mean



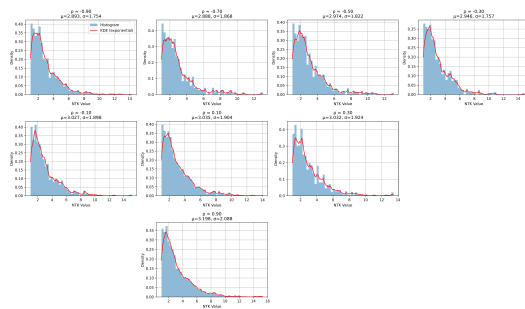
- Mean values alone are insufficient.
- Rank affects concentration across several statistics.
- The dependence on β remains visible empirically.
- The distribution becomes more regular as rank grows.

Distributional View of the Kernel

Log Log KSD and Tail Analysis (rank=1, $\beta=0.1$)

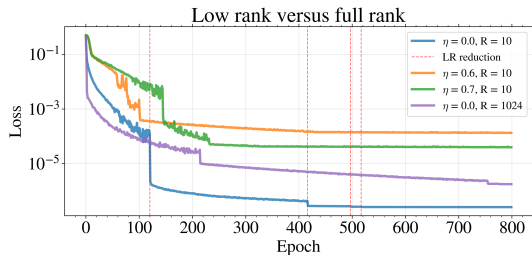


NTK Distributions by β (rank=1, $\beta=0.1$)

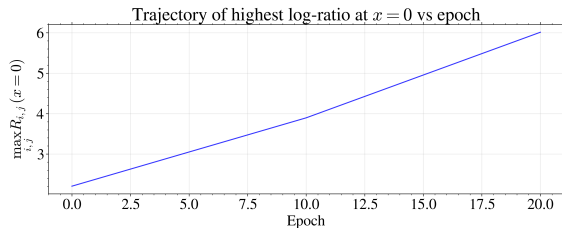


- Mean behavior hides relevant instability.
- The full distribution reveals heavy tails and sensitivity.
- Geometry changes the location of the mass.
- This helps explain unstable initializations.

Feature Learning and Landscape Dynamics

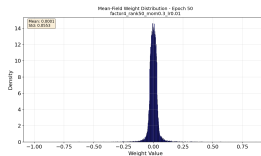
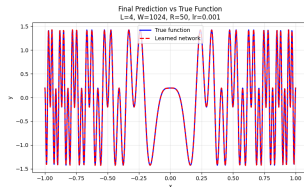
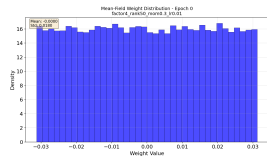
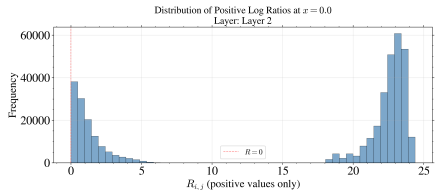


rank $r = 10$ vs full-rank ($r = 1024$) on $y = \cos(2\pi x)$, $x \in [-1, 1]$, 3 layers, width 1024, $n = 5000$, batch=4, SGD lr 0.01 with rec (momentum $\eta \in \{0, 0.3, 0.6, 0.7\}$) and one full-rank ($\eta = 0$). Red bars: SGD lr reductions for $\eta = 0$.



- Training is not only smooth kernel descent.
- Plateaus and sharp drops are clearly visible.
- Channel specialization appears during optimization.
- Log-ratios connect these effects to theory.

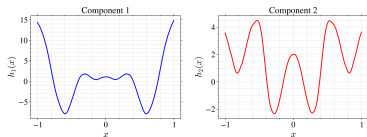
Representation Learning Through Training



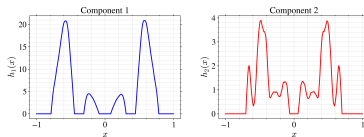
- Observables, learned functions, and mean-field motion agree.
- The representation moves substantially during training.
- This motion improves the final predictor.
- The regime is not purely lazy.

Internal Features Are Structured, Not Arbitrary

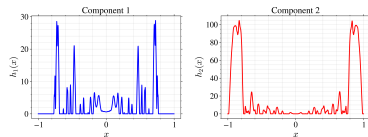
Partial functions – Layer 5
freq=(144,48) rank=15 fixWb=True



Partial functions – Layer 8
freq=(144,48) rank=15 fixWb=True



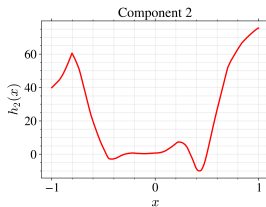
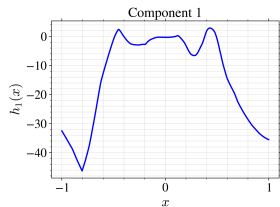
Partial functions – Layer 16
freq=(144,48) rank=15 fixWb=True



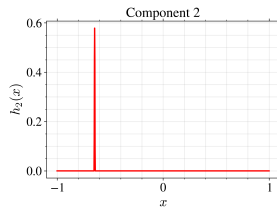
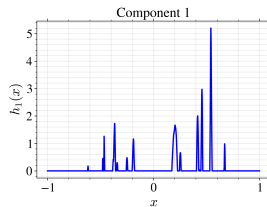
- Internal channels do not learn arbitrary shapes.
- They localize and sharpen useful patterns.
- Depth increases oscillation and hierarchy.
- The architecture organizes feature learning.

Low Rank Also Regularizes Geometry

Partial functions – Layer 5
freq=(144,48) rank=20 fixWb=False



Partial functions – Layer 16
freq=(144,48) rank=20 fixWb=False



- Removing the low-rank structure degrades the internal geometry.
- Symmetry breaks more easily.
- Rank is not only a parameter-saving device.
- It also acts as a geometric regularizer.

Theory

- Architecture can be analyzed through explicit control parameters.
- Depth, rank, and geometry appear in closed-form statements.

Practice

- The theoretical quantities are visible in numerical experiments.

Program

- This suggests a broader program for architecture-aware optimization.

Conclusion

Takeaway 1

- Deep optimization is hard. Architecture hits many levels at once.

Takeaway 2

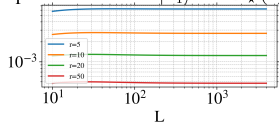
- Low rank: tractable model for those joint effects.

Takeaway 3

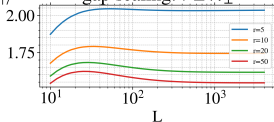
- Kernels, dynamics, experiments: one consistent story.

Appendix: Additional Figure Gallery I

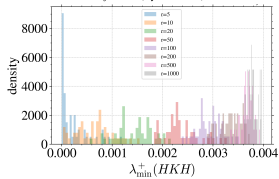
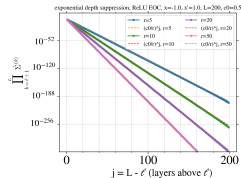
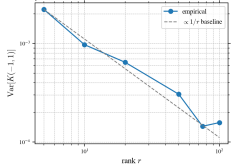
spike saturation: $L |\lambda_1/n - \Theta_\star(r)|/n$



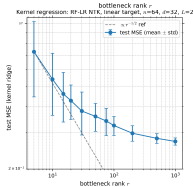
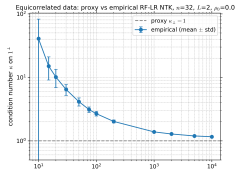
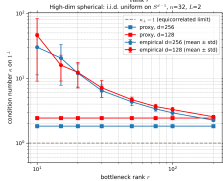
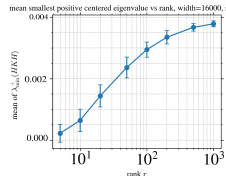
gap scaling: $r L \lambda_1$



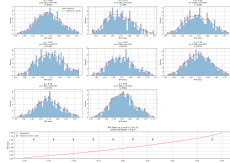
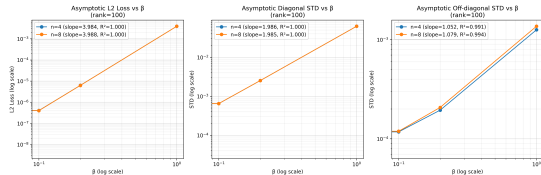
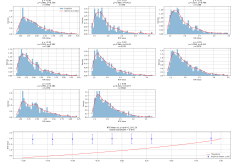
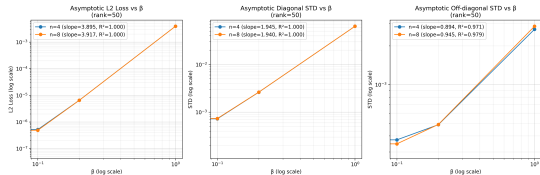
3-layer (2-ReLU) low-rank NTK variance, width=16000, trials=100



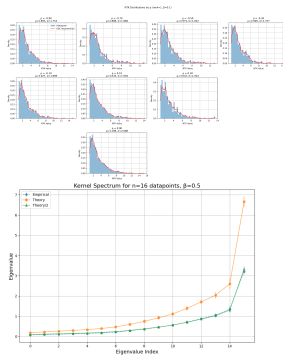
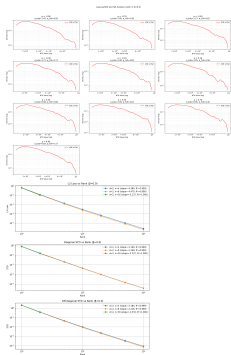
Appendix: Additional Figure Gallery II



Appendix: MMNN NTK Gallery I

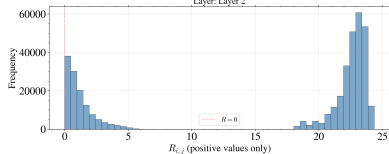


Appendix: MMNN NTK Gallery II

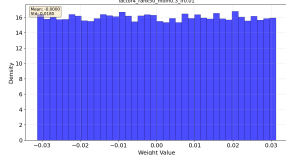


Appendix: Feature-Learning Gallery I

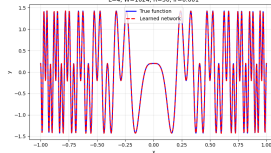
Distribution of Positive Log Ratios at $x = 0.0$
Layer: Layer 2



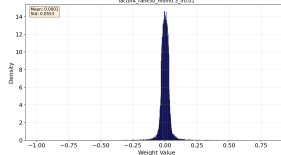
Mean-Field Weight Distribution - Epoch 0
Factor4, gain50, mixed 3.00.85



Final Prediction vs True Function
 $L=4, W=1024, R=50, lr=0.001$



Mean-Field Weight Distribution - Epoch 50
Factor4, gain50, mixed 3.00.85



Appendix: NTK Background Gallery

